

GOONDIWINDI, Australia

WMO: 945300

Lat: 28.521S

Lon: 150.326E

Elev: 219

StdP: 98.73

Time Zone: 10.00 (E10)

Period: 90-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 1.5	(c) 3.0	(d) -5.5	(e) 2.4	(f) 17.8	(g) -3.4	(h) 2.9	(i) 15.6	(j) 9.6	(k) 16.4	(l) 8.5	(m) 16.1	(n) 0.2	(o) 230	(p) 0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.6	37.4	20.8	35.7	20.8	34.2	20.5	24.0	30.7	23.4	30.0	22.7	29.4	3.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.6	17.7	25.9	21.6	16.7	25.8	20.8	15.9	25.7	73.4	30.5	70.8	29.9	68.3	29.3	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.6	7.4	6.4		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
				DB												
				WB	-1.2	26.0	1.4	1.7	-2.2	27.2	-3.0	28.2	-3.8	29.1	-4.8	30.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.8	26.8	26.1	23.9	20.1	16.3	12.7	11.7	13.3	16.7	20.3	23.9	25.7
	DBStd	6.06	2.56	2.96	2.48	2.76	2.57	3.02	2.86	3.18	3.53	3.00	3.37	2.80
	HDD10.0	29	0	0	0	0	0	8	14	7	1	0	0	0
	HDD18.3	707	0	0	1	13	73	169	206	159	70	14	1	0
	CDD10.0	3589	521	450	431	303	194	88	67	109	203	319	417	486
	CDD18.3	1224	263	217	172	66	9	1	0	3	22	76	168	228
	CDH23.3	13243	3016	2447	1676	592	59	2	2	49	299	756	1874	2472
	CDH26.7	5919	1536	1190	662	126	3	0	0	8	84	244	879	1186
(14) Wind	WSAvg	2.1	2.5	2.1	1.8	1.6	1.4	1.5	1.7	2.0	2.4	2.7	3.0	2.6
(15) Precipitation	PrecAvg	611	83	64	56	43	49	23	38	34	35	49	64	72
	PrecMax	931	235	282	167	272	195	73	133	146	114	146	175	234
	PrecMin	341	4	5	0	0	0	2	0	0	0	5	1	1
	PrecStd	136	57	58	53	58	38	18	31	27	27	36	44	46
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	38.6	36.7	31.4	27.1	23.5	23.3	27.3	32.1	33.7	39.0	38.1
		MCWB	21.5	22.6	20.3	18.7	17.2	15.7	15.2	15.6	17.5	18.0	18.4	20.3
	2%	DB	37.2	35.9	33.5	29.4	25.1	22.1	21.2	24.2	29.5	31.3	35.9	36.1
		MCWB	21.5	21.7	20.2	18.0	16.3	15.2	13.8	14.1	16.8	17.5	20.1	20.4
	5%	DB	35.2	34.3	31.7	27.9	23.7	20.4	19.9	22.2	26.9	29.6	33.4	34.2
		MCWB	21.1	21.4	19.8	17.2	15.5	14.0	13.2	13.2	15.4	16.9	18.5	20.3
	10%	DB	33.5	32.6	30.3	26.7	22.4	18.9	18.3	20.6	24.7	27.9	31.1	32.4
		MCWB	20.7	20.9	19.3	16.8	15.1	13.6	12.3	12.7	14.9	16.5	18.1	20.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.7	25.0	24.0	21.3	19.7	17.5	17.4	16.9	20.5	21.0	23.0	24.4
		MCDB	31.7	32.9	28.8	26.7	23.5	21.0	20.7	23.7	28.4	28.1	28.5	31.4
	2%	WB	23.7	24.2	22.9	20.2	18.4	16.4	15.6	15.7	18.5	19.6	21.7	23.2
		MCDB	31.2	30.7	28.6	26.1	21.5	19.7	18.8	20.9	24.9	26.7	28.6	31.1
	5%	WB	23.1	23.5	22.1	19.3	17.5	15.4	14.3	14.8	17.1	18.6	20.7	22.3
		MCDB	30.3	29.7	27.4	24.9	21.1	19.0	18.0	20.3	23.7	25.3	28.1	29.6
	10%	WB	22.3	22.8	21.2	18.2	16.5	14.3	13.3	13.8	16.1	17.8	20.0	21.5
		MCDB	29.5	29.1	26.9	24.0	20.9	18.1	17.1	19.2	22.8	24.5	27.5	28.7
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	11.7	12.4	13.1	12.2	12.0	12.3	14.3	14.1	13.2	12.8	11.9
		MCDBR	14.1	13.3	13.9	14.4	13.8	13.0	14.1	15.8	17.0	15.7	15.4	14.0
	5% WB	MCWBR	4.1	4.0	4.6	5.5	6.0	6.7	7.3	7.4	6.7	5.4	5.1	4.5
		MCDWR	11.0	11.3	10.7	11.5	8.6	10.3	10.3	13.6	12.5	12.4	11.6	11.7
(40) Clear-Sky Solar Irradiance	taub	taub	0.362	0.359	0.334	0.318	0.299	0.292	0.286	0.299	0.332	0.355	0.372	0.369
		taud	2.524	2.543	2.612	2.612	2.613	2.645	2.638	2.571	2.472	2.442	2.443	2.484
	Ebn at Noon	Ebn at Noon	979	965	960	928	903	888	909	935	945	958	963	975
		Edh at Noon	112	107	95	86	77	71	74	88	107	117	121	117
(44) All-Sky Solar Radiation	RadAvg	7.39	6.78	5.86	4.94	3.90	3.21	3.60	4.61	5.72	6.58	7.14	7.36	
(45) Radiation	RadStd	0.47	0.53	0.34	0.22	0.18	0.26	0.24	0.28	0.41	0.50	0.63	0.59	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+0.50	N/A	N/A	N/A	+1.06	N/A	N/A	N/A	N/A	+199	+141

Nomenclature: See separate page